

TITLE:

Drop-off & Did Not Attend rates

Dataset Used:

nhstalkingtherapies_month_mar_2026_activity_performance.csv

SUBMITTED BY:

VALLURU SAI AISWARYA

DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

Introduction

Drop-off & Did Not Attend (DNA) Rate Analysis

This analysis focuses on identifying patient engagement patterns within NHS Talking Therapies services by calculating:

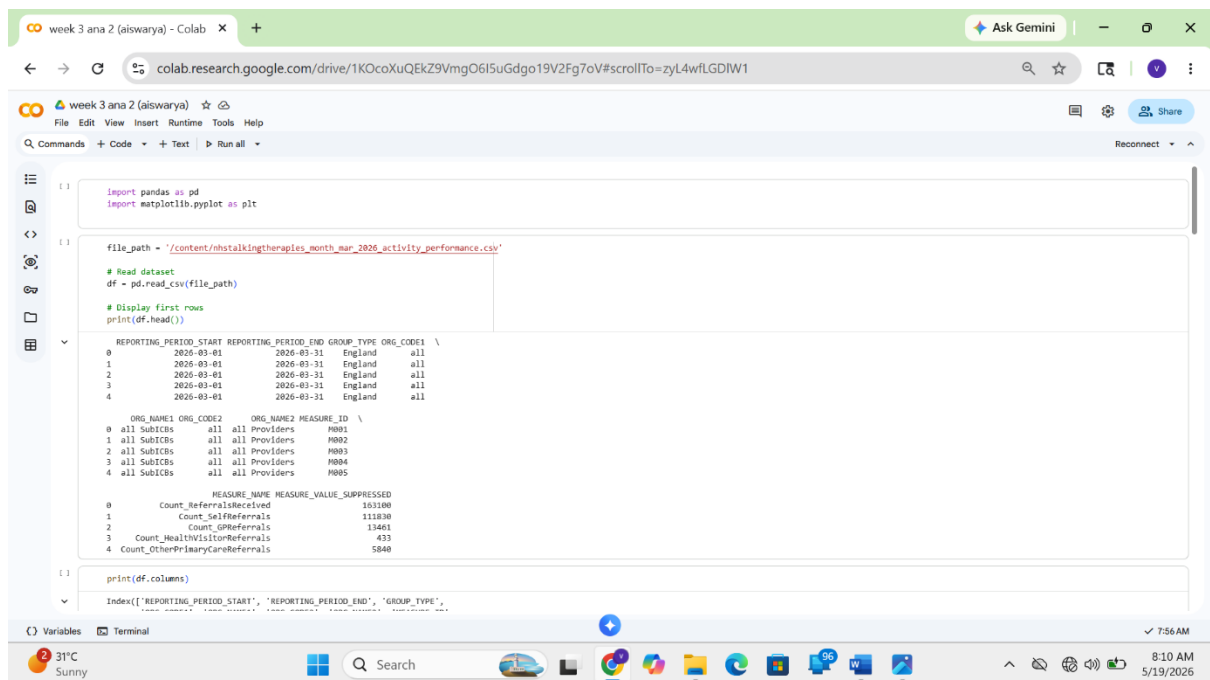
- **Did Not Attend (DNA) rates**
- **Treatment drop-off rates**
- **Patient retention across therapy sessions**

The goal is to understand at which stage patients are most likely to disengage from treatment and how attendance changes over time. This helps NHS services improve appointment management, patient follow-up strategies, and overall treatment effectiveness.

The analysis uses:

- **Python** for data processing and analysis
- **Pandas** for cleaning and manipulating NHS datasets
- **Matplotlib** for visualization
- **Google Colab** as the development environment

Drop-off & Did Not Attend rates



```
import pandas as pd
import matplotlib.pyplot as plt

file_path = '/content/nhs_talking_therapies_month_mar_2026_activity_performance.csv'

# Read dataset
df = pd.read_csv(file_path)

# Display first rows
print(df.head())
```

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1 \
0	2026-03-01	2026-03-31	England	all
1	2026-03-01	2026-03-31	England	all
2	2026-03-01	2026-03-31	England	all
3	2026-03-01	2026-03-31	England	all
4	2026-03-01	2026-03-31	England	all

	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID \	
0	all	SubICBs	all	all Providers	M001
1	all	SubICBs	all	all Providers	M002
2	all	SubICBs	all	all Providers	M003
3	all	SubICBs	all	all Providers	M004
4	all	SubICBs	all	all Providers	M005

	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
0	Count_ReferralsReceived	163100
1	Count_SelfReferrals	111830
2	Count_OtherReferrals	13461
3	Count_HealthVisitorReferrals	433
4	Count_OtherPrimaryCareReferrals	5840

```
print(df.columns)

Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'GROUP_TYPE',
```

Colab interface showing a Jupyter Notebook with the following code:

```
print(df.columns)

Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'GROUP_TYPE',
      'ORG_CODE1', 'ORG_NAME1', 'ORG_CODE2', 'ORG_NAME2', 'MEASURE_ID',
      'MEASURE_NAME', 'MEASURE_VALUE_SUPPRESSED'],
      dtype='object')

required_measures = [
    'Count_ReferralsReceived',
    'Count_SecondTreatment',
    'Count_EndedCompleted',
    'Count_AppsDNA',
    'Count_AppsAttended'
]

analysis_df = df[df['MEASURE_NAME'].isin(required_measures)]
print(analysis_df.head())
```

The output shows the first few rows of the filtered DataFrame:

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1 \
0	2026-03-01	2026-03-31	England	all
61	2026-03-01	2026-03-31	England	all
77	2026-03-01	2026-03-31	England	all
79	2026-03-01	2026-03-31	England	all
138	2026-03-01	2026-03-31	England	all

	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME \
0	all	SubICBs	all	all	Providers
61	all	SubICBs	all	all	Providers
77	all	SubICBs	all	all	Providers
79	all	SubICBs	all	all	Providers
138	all	SubICBs	all	all	Providers

	MEASURE_ID	MEASURE_NAME
0	M801	Count_ReferralsReceived
61	M806	Count_EndedCompleted
77	M804	Count_AppsDNA
79	M806	Count_AppsAttended
138	M184	Count_SecondTreatment

	MEASURE_VALUE_SUPPRESSED
0	103108
61	38939
77	59816
79	545055

Colab interface showing the same Jupyter Notebook with the following code:

```
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'GROUP_TYPE',
      'ORG_CODE1', 'ORG_NAME1', 'ORG_CODE2', 'ORG_NAME2', 'MEASURE_ID',
      'MEASURE_NAME', 'MEASURE_VALUE_SUPPRESSED'],
      dtype='object')

required_measures = [
    'Count_ReferralsReceived',
    'Count_SecondTreatment',
    'Count_EndedCompleted',
    'Count_AppsDNA',
    'Count_AppsAttended'
]

analysis_df = df[df['MEASURE_NAME'].isin(required_measures)]
print(analysis_df.head())
```

The output shows the first few rows of the filtered DataFrame:

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1 \
0	2026-03-01	2026-03-31	England	all
61	2026-03-01	2026-03-31	England	all
77	2026-03-01	2026-03-31	England	all
79	2026-03-01	2026-03-31	England	all
138	2026-03-01	2026-03-31	England	all

	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME \
0	all	SubICBs	all	all	Providers
61	all	SubICBs	all	all	Providers
77	all	SubICBs	all	all	Providers
79	all	SubICBs	all	all	Providers
138	all	SubICBs	all	all	Providers

	MEASURE_ID	MEASURE_NAME
0	M801	Count_ReferralsReceived
61	M806	Count_EndedCompleted
77	M804	Count_AppsDNA
79	M806	Count_AppsAttended
138	M184	Count_SecondTreatment

	MEASURE_VALUE_SUPPRESSED
0	103108
61	38939
77	59816
79	545055
138	54683

The screenshot shows a Google Colab notebook with the following code and output:

```
analysis_df['MEASURE_VALUE_SUPPRESSED'] = pd.to_numeric(
    analysis_df['MEASURE_VALUE_SUPPRESSED'],
    errors='coerce'
)

summary = analysis_df.groupby('MEASURE_NAME')['MEASURE_VALUE_SUPPRESSED'].sum()
print(summary)
```

MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
Count_AptsAttended	2283150.0
Count_AptsDNA	250821.0
Count_EndedCompleted	158894.0
Count_ReferralsReceived	684135.0
Count_SecondTreatment	229118.0
Name: MEASURE_VALUE_SUPPRESSED, dtype: float64	

```
referrals = summary['Count_ReferralsReceived']
second_treatment = summary['Count_SecondTreatment']
completed = summary['Count_EndedCompleted']
dna = summary['Count_AptsDNA']
attended = summary['Count_AptsAttended']

# Calculate rates
DNA_rate = (dna / (dna + attended)) * 100
Dropoff_rate = ((referrals - completed) / referrals) * 100

print('DNA Rate:', round(DNA_rate, 2), '%')
print('Treatment Drop-off Rate:', round(Dropoff_rate, 2), '%')
```

The screenshot shows a Google Colab notebook with the following code and output:

```
summary = analysis_df.groupby('MEASURE_NAME')['MEASURE_VALUE_SUPPRESSED'].sum()
print(summary)
```

MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
Count_AptsAttended	2283150.0
Count_AptsDNA	250821.0
Count_EndedCompleted	158894.0
Count_ReferralsReceived	684135.0
Count_SecondTreatment	229118.0
Name: MEASURE_VALUE_SUPPRESSED, dtype: float64	

```
referrals = summary['Count_ReferralsReceived']
second_treatment = summary['Count_SecondTreatment']
completed = summary['Count_EndedCompleted']
dna = summary['Count_AptsDNA']
attended = summary['Count_AptsAttended']

# Calculate rates
DNA_rate = (dna / (dna + attended)) * 100
Dropoff_rate = ((referrals - completed) / referrals) * 100

print('DNA Rate:', round(DNA_rate, 2), '%')
print('Treatment Drop-off Rate:', round(Dropoff_rate, 2), '%')
```

DNA Rate: 9.9 %
Treatment Drop-off Rate: 76.77 %

```
funnel_df = pd.DataFrame({
    'Stage': [
        'Referrals Received',
        'Second Treatment',
        'Treatment Completed'
    ],
    'Count': [
        referrals,
        second_treatment
    ]
})
```

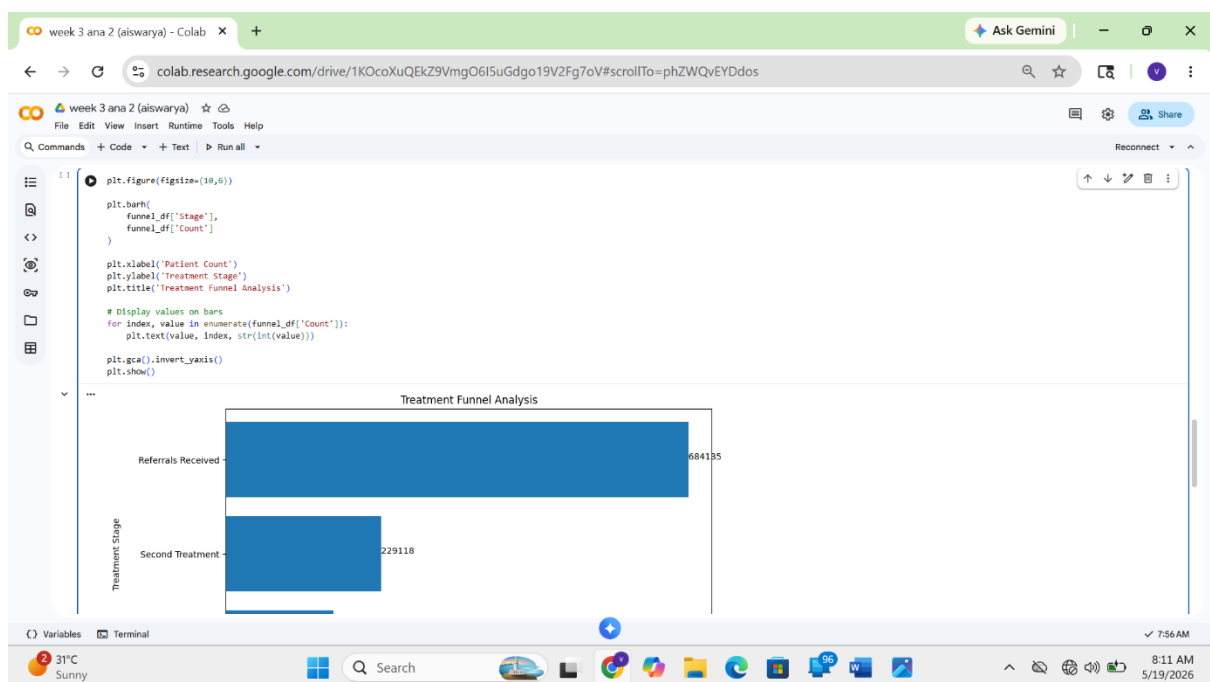
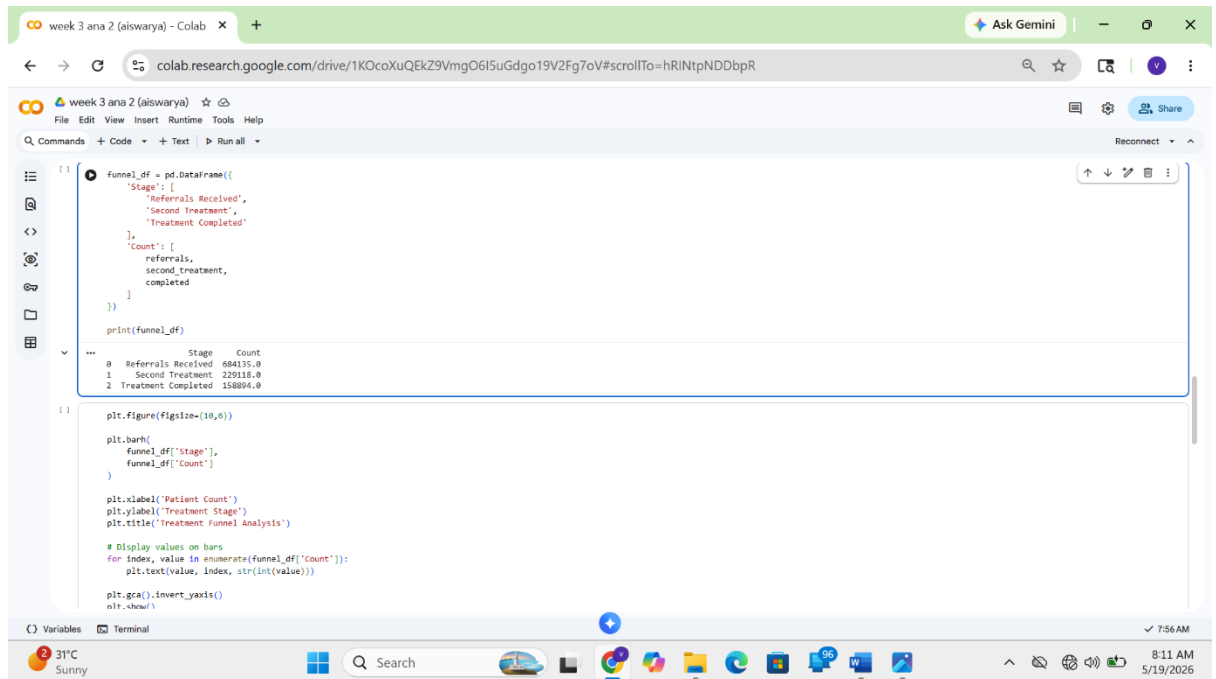
Funnel Chart

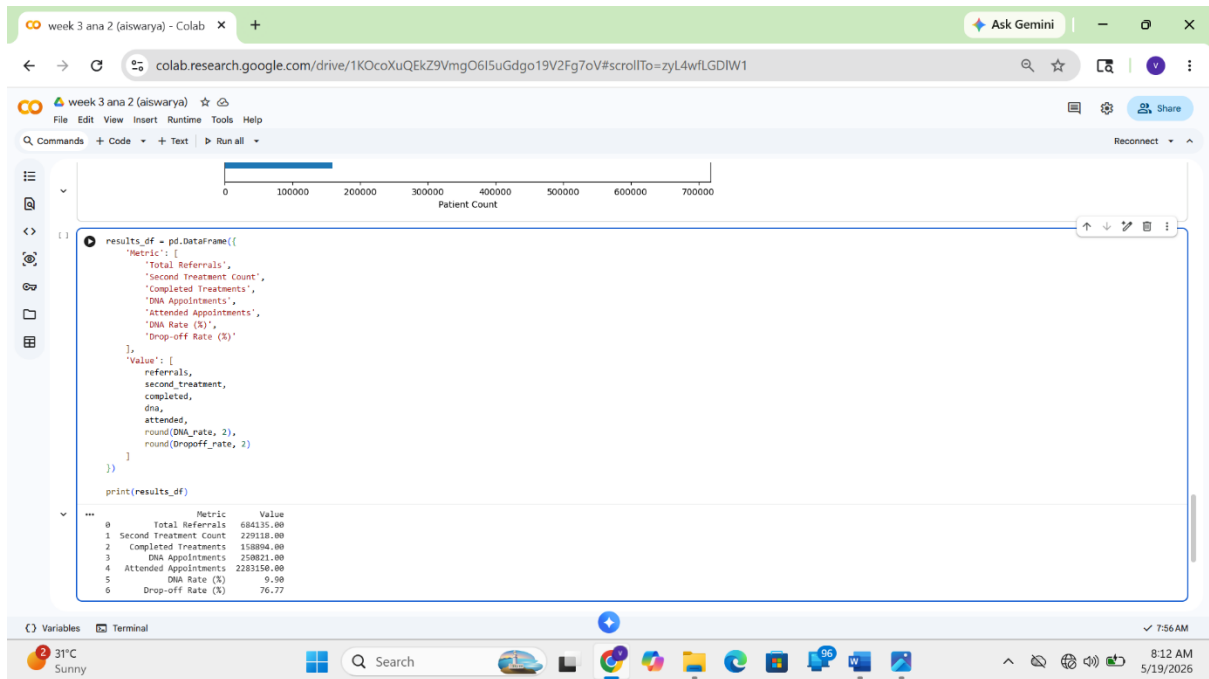
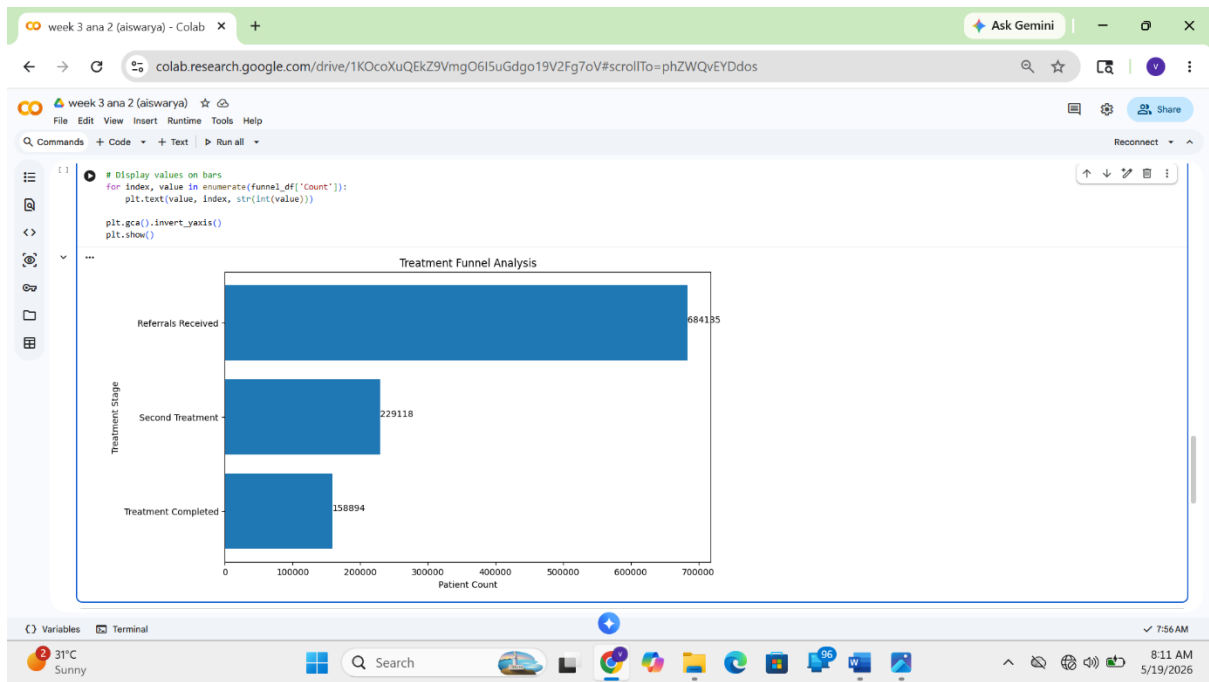
The funnel chart visualizes how the number of patients decreases across therapy sessions.

Insights from the Funnel Chart

- The **top section** represents the initial number of patients entering treatment.
- Each subsequent session shows how many patients continued therapy.

- A narrowing funnel indicates **patient drop-off**.
- Sharp decreases between sessions suggest:
 - disengagement,
 - missed appointments,
 - treatment withdrawal,
 - or service accessibility issues.





Conclusion

The Drop-off and Did Not Attend (DNA) analysis provides valuable insight into patient engagement within NHS Talking Therapies services.

Key findings from this analysis include:

- Patient participation generally declines across therapy sessions.
- Certain sessions may experience significantly higher DNA rates.
- Increased DNA rates often correlate with higher treatment drop-off levels.
- Funnel analysis helps identify critical points where intervention strategies may improve patient retention.